WhatsApp Multi-Agent Architecture - Complete Guide

Date: November 16, 2025
Base Workflow: `CUSTOMER-WHATSAPP-001: Donna AI - Main Agent` (ID: `86WHKNpj09tV9j1d`)
Status: 📋 Architecture Plan

🎯 OVERVIEW

Goal: Create a scalable WhatsApp agent system where:

1. ✅ Multiple agents for different purposes (customer support, sales, technical, etc.)
2. ✅ Website chatbots (embedded WhatsApp widgets)
3. ✅ n8n workflows with WhatsApp integration
4. ✅ Per-agent conversation isolation
5. ✅ Easy duplication and customization

🏗️ ARCHITECTURE

Current Setup (Donna AI - Single Agent)

```
WhatsApp Number: +1 214-436-2102
    ↓
WAHA Session: "default"
    ↓
Single Workflow: CUSTOMER-WHATSAPP-001
    ↓
Single Agent: Donna AI (Kitchen Design)
```

Target Architecture (Multi-Agent System)

```
WhatsApp Number: +1 214-436-2102
    ↓
WAHA Session: "default"
    ↓
Router Workflow: INT-WHATSAPP-ROUTER-001
    ├─→ Agent 1: CUSTOMER-WHATSAPP-001 (Donna AI - Kitchen Design)
    ├─→ Agent 2: CUSTOMER-WHATSAPP-002 (Sales Agent)
    ├─→ Agent 3: CUSTOMER-WHATSAPP-003 (Support Agent)
    ├─→ Agent 4: INT-WHATSAPP-WEBSITE-001 (Website Chatbot)
    └─→ Agent 5: INT-WHATSAPP-N8N-001 (Generic n8n Workflow Trigger)
```

📋 AGENT TYPES

1. Customer-Specific Agents (`CUSTOMER-WHATSAPP-*`)

Purpose: Dedicated agents for specific customers/projects

Examples:

- `CUSTOMER-WHATSAPP-001` : Donna AI (Kitchen Design for Dima)
- `CUSTOMER-WHATSAPP-002` : Tax4Us Support Agent
- `CUSTOMER-WHATSAPP-003` : Shelly Insurance Agent

Characteristics:

- Custom knowledge base per agent
- Customer-specific system prompts
- Isolated conversation memory
- Customer branding/personality

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2. Internal Agents (`INT-WHATSAPP-*`)

Purpose: Rensto internal operations

Examples:

- `INT-WHATSAPP-ROUTER-001` : Routes messages to correct agent
- `INT-WHATSAPP-WEBSITE-001` : Website chatbot (public-facing)
- `INT-WHATSAPP-SALES-001` : Sales qualification agent
- `INT-WHATSAPP-SUPPORT-001` : Customer support agent

Characteristics:

- Rensto branding
- General knowledge base
- Multi-purpose functionality

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3. Website Chatbots (`INT-WHATSAPP-WEBSITE-*`)

Purpose: Embedded WhatsApp widgets on website

Examples:

- Homepage chatbot
- Service page chatbots (Marketplace, Subscriptions, etc.)
- Niche page chatbots

Characteristics:

- Public-facing
- Lead qualification
- FAQ responses
- Handoff to human when needed

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4. n8n Workflow Triggers (`INT-WHATSAPP-N8N-*`)

Purpose: Generic WhatsApp integration for any n8n workflow

Examples:

- Lead generation workflows
- Form submissions via WhatsApp
- Notification workflows
- Data collection workflows

Characteristics:

- Generic, reusable
- Easy to connect to existing workflows
- No AI agent needed (just message routing)

🔄 ROUTING STRATEGY

Option 1: Keyword-Based Routing (Simple)

How It Works:

- Router workflow checks first message for keywords
- Routes to appropriate agent based on keywords

Keywords:

- "sales", "buy", "pricing" → Sales Agent
- "support", "help", "issue" → Support Agent
- "donna", "kitchen" → Donna AI
- Default → Website Chatbot

Pros: Simple, fast
Cons: Requires specific keywords, less flexible

Option 2: Phone Number Routing (Recommended)

How It Works:

- Router checks sender's phone number
- Routes to agent based on phone number mapping (Airtable/Boost.space)

Mapping Table (Airtable/Boost.space):

Phone Number	Agent Workflow ID	Agent Name	Purpose
+12144362102	86WHKNpj09tV9j1d	Donna AI	Kitchen Design
+1234567890	INT-WHATSAPP-SALES-001	Sales Agent	Lead Qualification
Default	INT-WHATSAPP-WEBSITE-001	Website Chatbot	General Inquiries

Pros: Flexible, per-user routing, easy to manage
Cons: Requires phone number database

Option 3: AI-Powered Routing (Advanced)

How It Works:

- Router uses AI to analyze first message
- Determines intent and routes to best agent

Pros: Most flexible, intelligent routing
Cons: Slower, requires AI call, more complex

🚀 IMPLEMENTATION PLAN

Phase 1: Test Current Workflow ✅ DO FIRST

1. **Test Donna AI Workflow**:
   - Send WhatsApp message to `+1 214-436-2102`
   - Verify workflow triggers
   - Check execution in n8n
   - Fix any issues
2. **Verify WAHA Connection**:
   - Check WAHA dashboard: `http://173.254.201.134:3000/dashboard`
   - Verify webhook is configured

- Test webhook manually if needed

Phase 2: Create Router Workflow

Workflow: `INT-WHATSAPP-ROUTER-001`

Structure:

```
WAHA Trigger
    ↓
Get Phone Number
    ↓
Lookup Agent (Airtable/Boost.space)
    ↓
Route to Agent (Switch Node)
    ├─→ Agent 1 (Sub-Workflow)
    ├─→ Agent 2 (Sub-Workflow)
    ├─→ Agent 3 (Sub-Workflow)
    └─→ Default Agent (Sub-Workflow)
```

Nodes:

1. **WAHA Trigger**: Receives all WhatsApp messages
2. **Extract Phone**: Get sender's phone number
3. **Lookup Agent**: Query Airtable/Boost.space for agent mapping
4. **Switch Node**: Route to correct agent workflow
5. **Sub-Workflow Nodes**: Call specific agent workflows

Phase 3: Duplicate Base Workflow

Template: `CUSTOMER-WHATSAPP-001: Donna AI`

Duplication Steps:

1. Duplicate workflow in n8n
2. Update workflow name (e.g., `CUSTOMER-WHATSAPP-002: Sales Agent`)
3. Update system prompt for new agent
4. Update knowledge base/store name
5. Update memory session key (if needed)
6. Test new agent

Quick Duplicate Script:

- Use n8n MCP tools to duplicate workflow
- Update specific fields
- Activate new workflow

Phase 4: Create Agent Templates

Template Categories:

1. **RAG Agent Template** (Like Donna AI):
    - WAHA Trigger
    - Message Filter
    - Voice/Text Router
    - AI Agent with RAG
    - TTS

    - Send Response

2. **Simple Chatbot Template**:
    - WAHA Trigger
    - AI Agent (no RAG)
    - Send Response
3. **Workflow Trigger Template**:
    - WAHA Trigger
    - Extract Data
    - Call Another Workflow
    - Send Confirmation

Phase 5: Website Integration

Option A: WhatsApp Widget (Recommended)

Implementation:

1. Add WhatsApp widget to website
2. Widget opens WhatsApp chat with pre-filled message
3. User sends message → Router → Website Chatbot Agent
4. Agent responds with FAQ or qualification questions

Widget Code (for Vercel/Next.js):

```
// components/WhatsAppWidget.tsx
export const WhatsAppWidget = () => {
  const phoneNumber = "+12144362102";
  const message = encodeURIComponent("Hello! I'm interested in Rensto's services.");
  const whatsappUrl = `https://wa.me/${phoneNumber}?text=${message}`;

  return (
    <a href={whatsappUrl} target="_blank" rel="noopener noreferrer">
      <button>💬 Chat on WhatsApp</button>
    </a>
  );
};
```

Option B: Embedded Chat (Advanced)

- Use WhatsApp Business API
- Embed chat widget directly on page
- Requires WhatsApp Business Account

📊 AGENT CONFIGURATION

Agent Configuration Table (Airtable/Boost.space)

Table: `WhatsApp Agents`

Field	Type	Description
agent_id	Text	Unique ID (e.g., "donna-ai")
workflow_id	Text	n8n workflow ID

agent_name	Text	Display name (e.g., "Donna AI")
purpose	Text	Purpose description
system_prompt	Long Text	AI system prompt
knowledge_store	Text	Gemini File Search store name
phone_numbers	Multiple Selects	Authorized phone numbers
is_active	Checkbox	Active status
created_date	Date	Creation date

🔧 DUPLICATION PROCESS

Step-by-Step: Duplicate Donna AI to New Agent

1. **Duplicate Workflow**:

```
# Use n8n MCP tool
n8n_get_workflow({ id: "86WHKNpj09tV9j1d" })
# Save as new workflow
n8n_create_workflow({ name: "CUSTOMER-WHATSAPP-002: Sales Agent", ... })
```

2. **Update Workflow Name**:
   - Change: `CUSTOMER-WHATSAPP-001: Donna AI - Main Agent`
   - To: `CUSTOMER-WHATSAPP-002: Sales Agent`
3. **Update System Prompt**:
   - Find "Donna AI Agent" node
   - Update system message for sales context
4. **Update Knowledge Store** (if using RAG):
   - Change: `fileSearchStores/donna-kitchen-knowledge`
   - To: `fileSearchStores/sales-knowledge` (or create new store)
5. **Update Memory Session Key** (if needed):
   - Keep per-phone isolation: `={{ $('WAHA Trigger').item.json.payload.from }}`
6. **Update Agent Name in Responses**:
   - Change "Donna" to new agent name in system prompt
7. **Test New Agent**:
   - Send test message
   - Verify routing works
   - Check responses

🎯 USE CASES

Use Case 1: Customer Support Agent

Purpose: Handle customer support inquiries

Configuration:

- **System Prompt**: "You are a helpful customer support agent for Rensto..."
- **Knowledge Base**: FAQ, troubleshooting guides
- **Routing**: Phone numbers from support tickets

Features:

- Ticket creation
- Escalation to human
- Knowledge base search

Use Case 2: Sales Qualification Agent

Purpose: Qualify leads and schedule consultations

Configuration:

- **System Prompt**: "You are a sales qualification agent..."
- **Knowledge Base**: Service offerings, pricing
- **Routing**: New website inquiries

Features:

- Lead qualification questions
- Service recommendation
- Calendar scheduling
- Handoff to sales team

Use Case 3: Website Chatbot

Purpose: Answer general website questions

Configuration:

- **System Prompt**: "You are a friendly chatbot for Rensto's website..."
- **Knowledge Base**: Website content, service descriptions
- **Routing**: Default for unknown phone numbers

Features:

- FAQ responses
- Service information
- Lead capture
- Handoff to sales/support

Use Case 4: n8n Workflow Trigger

Purpose: Trigger any n8n workflow via WhatsApp

Configuration:

- **No AI Agent**: Just message routing
- **Workflow**: Calls target workflow with message data

Features:

- Generic message forwarding
- Data extraction
- Workflow triggering
- Confirmation responses

📝 NEXT STEPS

1. ✅ **Test Current Workflow**: Verify Donna AI is working
2. ⏳ **Create Router Workflow**: Build `INT-WHATSAPP-ROUTER-001`
3. ⏳ **Create Agent Database**: Set up Airtable/Boost.space table
4. ⏳ **Duplicate First Agent**: Create Sales Agent as example
5. ⏳ **Add Website Widget**: Integrate WhatsApp widget on website
6. ⏳ **Document Duplication Process**: Create step-by-step guide

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🔗 REFERENCES

- **Base Workflow**: `http://173.254.201.134:5678/workflow/86WHKNpj09tV9j1d`
- **WAHA Dashboard**: `http://173.254.201.134:3000/dashboard`
- **n8n Executions**: `http://173.254.201.134:5678/executions`
- **Donna AI Docs**: `dima/DONNA_AI_*` files

Last Updated: November 16, 2025
Status: 📋 Architecture Plan - Ready for Implementation